

Name of the Program:	B.Tech 2022-23	Semester:	II
Course/Subject Name:	ENGINEERING MATHEMATICS II	Course/Subject Code:	MTH11502
Maximum Marks:	50	Time Duration:	180 Minutes
Total No. of Questions:	12	Total No of Pages:	2
(Any other information for the student may be mentioned here)	1. At top sheet, clearly mention Univ. Roll No., Enrolment No., Course Name & Code, Date of Exam. 2. All parts of a Question should be answered consecutively. Each Answer should start from a fresh page. 3. Assumptions made if any, should be stated clearly at the beginning of your answer.		

Group A

Answer All 5 Questions (5 x 2 = 10)

		BTL	COs
1	Define sequence of real numbers. Make use of limit to check the convergence of the sequence $\frac{n^2 - 2n}{3n^2 + n}$. [1+1]	L1, L3	CO1
2	Explain Dirichlet's conditions for the existence of a Fourier series.	L2	CO1
3	Show that the function $f(x, y) = e^{-x}(x \sin y - y \cos y)$ is a harmonic function.	L2	CO2
4	Define singular point of an analytic function and find the singular point of the function $f(z) = \frac{e^z}{z-2}$. [1+1]	L1	CO2
5	Examine the following differential equation to be an exact differential equation: $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$	L4	CO3

Group B

Answer All 5 Questions (5 x 4 = 20)

6	Apply ratio test to check the convergence of the following series: $\frac{1}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^4}{4\sqrt{3}} + \frac{x^6}{5\sqrt{4}} + \dots$	L3	CO1
7	Find the half-range Fourier sine series for the function $f(x) = e^x$ in the interval $0 < x < 1$.	L1	CO1
8	Find the conjugate harmonic $v(x, y)$ of $u(x, y) = 2x - x^3 + 3xy^2$ so that $f(z) = u(x, y) + iv(x, y)$ is an analytic function.	L2	CO2
9	Make use of line integral to find work done by the force $\vec{F} = xy\hat{i} + (y-x)\hat{j}$ in moving an object from (1, 1) to (2, 3) along the straight line $C: y = 2x - 1$.	L3	CO4
10(a)	Evaluate the contour integral $\oint_C \frac{e^z}{(z+1)(z+2)} dz$, where C is the circle $ z-1 = 4$.	L4	CO2

OR

10(b)	Prove that $\oint_C \frac{1}{z-a} dz = 2\pi i$, where C is any simple closed curve and $z = a$ is an interior point of C .	L4	CO2
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Group C

Answer All 2 Questions (2 x 10 = 20)

11(a)	(i) Solve the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x(x \sin x)$.	L3	CO3
	(ii) Apply the method of variation of parameter to solve the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x(x \sin x)$. [5+5]	L3	CO3

OR

11(b)	(i) Solve the following Cauchy-Euler's differential equation: $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = \log x$	L3	CO3
	(ii) Solve the following simultaneous differential equations: $\frac{dx}{dt} - y = t, \frac{dy}{dt} + x = 1$ [5+5]	L3	CO3

12(a)	Evaluate the line integral $\oint_C [(y^2 - 7y)dx + (2xy + 2x)dy]$ along the circle $C: x^2 + y^2 = 1$. Also, verify the integral value using Green's theorem.	L5	CO4
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OR

12(b)	(i) Apply Stoke's theorem to find the line integral $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = xz\hat{i} + xy\hat{j} + 3xz\hat{k}$ and C is the boundary of the portion of the plane $2x + y + z = 2$ in the first octant traversed counter clockwise.	L3	CO4
	(ii) Apply Gauss divergence theorem to find the surface integral $\iiint_S [(x^3 - yz)\hat{i} - 2x^2y\hat{j} + 2\hat{k}] \cdot \hat{n} dS$, where S denotes the surface of the cube in space bounded by the planes $x=0, x=a, y=0, y=a, z=0, z=a$. [5+5]	L3	CO